

CASH-SETTLED SWAPTIONS: HOW WRONG ARE WE?

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ABSTRACT. The pricing of the European cash-settled swaptions is analysed. The standard market formula results are compared to results obtained from different models. Significant discrepancies are observed, justifying the title.

1. INTRODUCTION

In EUR and GBP the interbank standard for swaptions settlement is *cash-settlement*. So by the *The Market is always Right* hypothesis, their prices are correct. This partly answers the question in the title. Nevertheless one would like to price both cash-settled and physical delivery swaptions in a coherent way. The standard saying for the Black prices is that it is *the wrong parameter in the wrong model to obtain the correct price*. In the cash-settled swaption we would like to say that it is *the wrong parameter in the wrong formula from the wrong model*. This strong statement is the justification for the provocative title that we will try to justify in this note.

The term *cash-settled swaption* is used for swaption settling in cash using a simplified annuity (also called *yield-settled*). This is by opposition to the USD market where the standard is *cash-settled* with full revaluation (also called *zero-coupon cash-settled*).

The cash settlement mechanism consists in taking note on expiry date of the swap fixing (the same used for CMS) and settling the option value by a cash amount paid on expiry date¹. For a receiver swaption, the amount is computed as the positive part of the difference between the strike K and the fixing S_θ multiplied by the cash annuity

$$C(S)(K - S_\theta)^+.$$

The *cash-annuity* is the present value of a basis point using the rate as an internal rate of return, i.e. for a N years swaption with m annual payments and a fixing S the annuity is²

$$C(S) = \sum_{i=1}^{mN} \frac{\frac{1}{m}}{(1 + \frac{1}{m}S)^i}.$$

With that pay-off, the cash-settled swaptions should be considered as exotic options with respect to delivery swaptions. As the payoff is a complex function of the swap fixing, it can also be considered as an exotic CMS options. In standard CMS replication arguments, the opposite is done, the CMS is replicated by cash-settled swaptions. So we have a mechanism to move from cash-settled swaptions to CMS but no direct way to price any of them.

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¹The actual payment take place two days latter in EUR. The details are handled later.

²Note that the sum can be written explicitly as $C(S) = 1/S \left(1 - \frac{1}{(1+S/m)^{mN}}\right)$.

The standard market formula for the cash-delivery option imitates the approach used for physical delivery and at some stage in the proof substitutes the cash delivery numeraire by the physical delivery one (the proof is recalled later). This *substitution* is not mathematically justified and is standard market practice. A standard market formula is presented in (Brigo and Mercurio, 2006, Section 6.7.2). Here we use a different version which is used more often in practice and described in Mercurio (2007) and Mercurio (2008).

This note analyses if such a substitution introduces a non-coherence between the cash and physical delivery swaptions. A question related to cash-settled swaptions was asked in Mercurio (2007) and Mercurio (2008). The question is related to the arbitrage free property of the market formula smile. The answer to that question is no, the market formula is not arbitrage free! In that sense this note is not required as it is already known that an arbitrage opportunity exists. In another sense this note is useful as one would like to know if the discrepancy between cash and physical settle swaptions appears also locally.

To achieve the goal to measure the error introduced we need some way to obtain the *exact* (at least precise enough) value. To our knowledge it is not possible to write the price of a cash-settled swaption as function of delivery swaptions prices or inversely in a model independent way. We use several standard models to price the cash-settle swaptions. The first one is a one factor Gaussian HJM model (extended Vasicek or Hull-White). In that approach the difference is non-negligible. Then we use a two factor Gaussian model (G2++) to check if the multi-factor curve movements have an impact on the difference. A priori, as in the cash-settled swaptions the discounting is done with the *wrong* rate, curve shape and non-parallel moves (through mean reversion or multi-factor) could have an impact. In the next section we analyze the prices in a Libor Market Model (LMM) with displaced diffusion. The flexibility of the model allows to calibrate to swaptions of all tenors and to the volatility skew. The impact of the multi-factor and skew features are analysed.

The general observation is that the difference between the market standard formula and the model prices are larger than acceptable. The market price is very often outside the range of the prices provided by the different models.

The tests are done by calibrating the model to physical delivery swaptions prices for which simple model price formulas exists. The prices of cash-settled swaptions are obtained from those calibrations through numerical procedures (integration or Monte Carlo). In practice the opposite should be done as the most liquid swaptions are the cash-settled ones. Had we done it the opposite way, the qualitative results, which are comparisons between different prices and not absolute prices, would have been the same. The calibration process would have been numerically longer and less stable.

In the Appendix we propose an approximated explicit formula for cash-settled swaptions in the Hull-White model.

2. GENERAL DESCRIPTION AND MARKET FORMULA

The analysis framework is a *multi-curves setting* as described in Henrard (2010a). There is one discounting curve denoted $P^D(s, t)$ and one forward curve $P^j(s, t)$ where j is the relevant Libor tenor.

The underlying swap has a start date t_0 , fixed leg payment dates $(t_i)_{1 \leq i \leq n}$ and floating leg payment dates $(\tilde{t}_i)_{1 \leq i \leq \tilde{n}}$. The strike (or coupon) is K and the accrual fraction for each fixed period is $(\delta_i)_{1 \leq i \leq n}$. The expiry date of the swaption is $\theta \leq t_0$. The analysis is done for a receiver swaption (the figures and final results are given for both receiver and payer swaptions).

The (delivery) *annuity* is

$$A_t = \sum_{i=1}^n \delta_i P^D(t, t_i).$$

The swap rate³. is

$$S_t = \frac{\sum_{i=1}^{\tilde{n}} P^D(t, \tilde{t}_i) \left(\frac{P^j(t, \tilde{t}_i)}{P^j(t, \tilde{t}_{i+1})} - 1 \right)}{A_t}.$$

The pay-off of a delivery swaption is given in θ by

$$\left(\sum_{i=0}^n \delta_i K P(\theta, t_i) - \sum_{i=1}^{\tilde{n}} P^D(\theta, \tilde{t}_i) \left(\frac{P^j(\theta, \tilde{t}_i)}{P^j(\theta, \tilde{t}_{i+1})} - 1 \right) \right)^+ = A_\theta (K - S_\theta)^+.$$

The (Black) market price of delivery swaptions is obtained in the following way. The generic price is $N_0 \mathbb{E}^N [N_\theta^{-1} A_\theta (K - S_\theta)^+]$ for a numeraire N and its associated measure. If A_t is chosen as numeraire, the rate S_t is a martingale (asset price/numeraire ratio) and one can chose as stochastic equation for S_t a geometric Brownian motion without drift

$$dS_t = \sigma S_t dW_t.$$

In the annuity numeraire, the price simplifies to $A_0 \mathbb{E}^A [(K - S_\theta)^+]$ which can be computed explicitly

$$A_0 \text{Black}(K, S_0, \sigma).$$

The pay-off of a cash-settled swaption is given in θ by

$$P^D(\theta, t_0) C(S_\theta) (K - S_\theta)^+.$$

The pay-offs are similar; the difference is in the annuity.

In the cash-settled case, the price is $N_0 \mathbb{E}^N [N_\theta^{-1} P^D(\theta, t_0) C(S_\theta) (K - S_\theta)^+]$. One choose $P^D(t, t_0) C(S_t)$ (a positive process) as numeraire and the prices become $P^D(0, t_0) C(S_0) \mathbb{E}^C [(K - S_\theta)^+]$. In that numeraire, S_θ is not necessarily a martingale (S_t is not an asset price/numeraire ratio anymore). The explicit process should stop here; nevertheless the market standard formula is to substitute C by A as numeraire and approximate the price by

$$P^D(0, t_0) C(S_0) \mathbb{E}^C [(K - S_\theta)^+] \simeq P^D(0, t_0) C(S_0) \mathbb{E}^A [(K - S_\theta)^+] = P^D(0, t_0) C(S_0) \text{Black}(K, S_0, \sigma).$$

Another way to obtain the same formula is to work directly in the physical annuity numeraire where the price is $A_0 \mathbb{E}^A [A_\theta^{-1} P^D(\theta, t_0) C(S_\theta) (K - S_\theta)^+]$. If one considers that the ratio $P^D(\theta, t_0) C(S_\theta) / A_\theta$ is a low variance variable and replaces it by its initial value (initial freeze technique)

$$\begin{aligned} A_0 \mathbb{E}^A [A_\theta^{-1} P^D(\theta, t_0) C(S_\theta) (K - S_\theta)^+] &\simeq A_0 \mathbb{E}^A [A_0^{-1} P^D(0, t_0) C(S_0) (K - S_\theta)^+] \\ &= P^D(0, t_0) C(S_0) \mathbb{E}^A [(K - S_\theta)^+]. \end{aligned}$$

None of the substitutions is theoretically justified. We verify if they are acceptable in practice.

Note that in the cash-settled swaptions case, there is no *put-call parity* in the sense that a long a receiver swaption and short a payer swaption is not a standard (forward) swap anymore. The result is another exotic product that could be called *cash swap* (but is not traded in the market). There is no model free price for that product. The analysis has to be done on both payer and receiver deals.

Note also that the ratio between the Black market price for physical delivery and the market standard formula for cash-settle is constant for all strikes: $A_0 / P(0, t_0) C(S_0)$, i.e. the market standard price is an annuity adjusted price that does not take into account the variation of the ratio.

In the delivery annuity numeraire, the exact expected value to compute for cash-settled options is $P(\theta, t_0) C(S_\theta) / A_\theta (K - S_\theta)^+$. A full term structure seems necessary to compute the value; a process for S_t only is not enough. In that sense there is no exact price in the Black world as the framework does not contain enough information to obtain an unambiguous price.

³For this note we ignore the difference between payment date and end fixing period date. In practice, due to week-end, they can be different by one day or two

3. MODELS

3.1. Extended Vasicek (Hull-White). We work with the model in its HJM version. The equations of the model in the numeraire measure associated to the cash-account N_t are

$$df(t, u) = \sigma(t, u)\nu(t, u)dt + \sigma(t, u)dW_t.$$

In the extended Vasicek model with time dependent volatility, we define

$$\sigma(s, t) = \eta(s) \exp(-a(t - s)) \quad \text{and} \quad \nu(s, t) = (1 - \exp(-a(t - s)))\eta(s)/a.$$

Let the (bond) volatilities on the period $(0, \theta)$ be denoted by

$$\alpha_i^2 = \int_0^\theta (\nu(s, t_i) - \nu(s, \theta))^2 ds.$$

If one select $P^D(., \theta)$ as numeraire, the discount factors can be written explicitly (see (Henrard, 2003, Lemma 3.1)) as

$$P^D(\theta, t_i) = \frac{P^D(0, t_i)}{P^D(0, \theta)} \exp\left(-\alpha_i X - \frac{1}{2}\alpha_i^2\right)$$

with the random variable X standard normally distributed.

The price of a physical delivery receiver swaption is given (see the above-mentioned paper) by

$$\sum_{i=0}^n c_i P^D(0, t_i) N(\kappa + \alpha_i)$$

where $(c_i)_{i=0, \bar{n}}$ are the swap cash-flow equivalent as described in Henrard (2010a) and κ is the (unique) solution of

$$\sum_{i=0}^n c_i P^D(0, t_i) \exp\left(-\frac{1}{2}\alpha_i^2 - \alpha_i \kappa\right) = 0.$$

The value of the cash-settled swaption is

$$(1) \quad P^D(0, t_0) E^\theta \left[\exp\left(-\alpha_0 X - \frac{1}{2}\alpha_0^2\right) C(S_\theta)(K - S_\theta)^+ \right].$$

3.2. G2++. In the cash-account numeraire, the forward rate equation is

$$df(t, u) = \nu(t, u) \cdot \sigma(t, u)dt + \sigma(t, u) \cdot dW_t$$

Let a_i ($i = 1, 2$) be the mean reversion and ρ the correlation. The following notation is used ($i = 1, 2$)

$$\tilde{\sigma}_i(s, u) = \eta_i(s) \exp(-a_i(u - s)).$$

Those functions satisfy the separability condition

$$\tilde{\sigma}_i(s, u) = g_i(s)h_i(u)$$

with $g_i(s) = \eta_i(s) \exp(a_i s)$ and $h_i(u) = \exp(-a_i u)$. The model volatilities are given by

$$\sigma(t, u) = (\tilde{\sigma}_1(t, u) + \tilde{\sigma}_2(t, u)\rho, \tilde{\sigma}_2(t, u)\sqrt{1 - \rho^2}),$$

$$\nu(t, u) = (\tilde{\nu}_1(t, u) + \tilde{\nu}_2(t, u)\rho, \tilde{\nu}_2(t, u)\sqrt{1 - \rho^2}).$$

In the $P^D(., \theta)$ numeraire, the discount factors can be written explicitly as

$$P^D(\theta, t_j) = \frac{P^D(0, t_j)}{P^D(0, \theta)} \exp\left(-\alpha_{1,j}X_1 - \alpha_{2,j}X_2 - \frac{1}{2}\tau_j^2\right)$$

with the random variables X_i standard normally distributed.

No explicit price formula for delivery swaption exists but very efficient approximation can be developed. One of such an approximation is described in Henrard (2010b).

3.3. Libor Market Model. The general description of the Libor Market model (LMM) can be found in the original paper [Brace et al. \(1997\)](#) or in books like [Rebonato \(2002\)](#). Here we use the version with displaced diffusion described below in the multi-curves framework. Here we use the deterministic spread hypothesis S_0 of [Henrard \(2010a\)](#).

The idea behind the Libor Market model (LMM) is to embed different Black-like equations for the forward (Libor) rate between standard dates $t_0 < \dots < t_n$ into a unique HJM model. The Libor rates L_s^i for deposits between t_i and t_{i+1} are defined by

$$1 + \delta_i L_s^i = \frac{P^j(s, t_i)}{P^j(s, t_{i+1})} = \beta^j(t_i, t_{i+1}) \frac{P^D(s, t_i)}{P^D(s, t_{i+1})} = \beta^j (1 + \delta_i D_s^i)$$

There is a direct (deterministic) link between the libor ratio L_s^i and the deposit forward rate D_s^i . The factors δ_i are the accrual factors or day count fractions and represent the fraction of the year spanned by the interval $[t_i, t_{i+1}]$ in the selected convention.

The equations underlying the original Libor Market Model are on the libor rates L_s^i , here we write them on the deposit

$$(2) \quad dD_t^j = \gamma_j(D_t^j, t).dW_t^{j+1}$$

in the probability space with numeraire $P^D(., t_{j+1})$. The γ_j ($0 \leq j \leq n-1$) are m -dimensional functions. For fundamental reasons not all such models are well-defined. Here it is supposed that the γ 's are such that the model described above is a well-defined HJM model.

The diplays diffusion model means that the local volatility functions are affine with

$$\gamma_j(D, t) = \gamma_j(t)(D + a_j)$$

with γ_j deterministic functions and a_j constants potentially different for each forward rate.

!Multi-curves framework explanation!

No explicit price formula for delivery swaption exists but very efficient approximation can be developed. One of such an approximation which is used here is described in [Henrard \(2010b\)](#).

4. MULTI MODELS ANALYSIS

The difference between physical and cash settled swaptions will depend strongly on the curve. Those differences are reproduced in Figure 1. The curves used have short term rate between 1 and 7 percent and slope (20Y rate minus over-night rate) between -0.50% to 4.00%. The underlying instruments are 5Yx10Y receiver swaptions with strike ATM+1.50%. For flat curves, the difference between the two types of swaptions is relatively small for all rate levels. For steep curves the difference can be important. It reaches 12 times the vega for the steepest (+4%) and lowest level (O/N at 1%) curves.

4.1. Data. For the first set of tests on the models we use EUR market data from 30 April 2010. The curves are the curves relevant for swaps with six months floating legs.

We analyze the cash-settle swaptions for three model types: extended Vasicek, G2++ and LMM. For each of them a range of parameters is used to see the price parameter dependency.

The main options used are 5Yx10Y swaptions, i.e. a five year option on a ten year rate. The option is chosen sufficiently long to see the volatility impact. The strike are regularly space from ATM-3% to ATM +4.5%. The forward is around 4.39%. For the smile description, we use a SABR approximate formula (see [Hagan et al. \(2002\)](#)) with realistic parameters.

For each test, the same approach is used. A certain number of parameters are selected (arbitrarily) and the others are calibrated to the price of delivery swaptions. The delivery swaption of same maturity and same strike is always part of the calibration basket. We analyze the price difference between the Black-like standard market formula and the model prices given by the different models.

As a reference we report a figure we call vega. This is a measure of sensitivity to the volatility level. In our case we use a one basis point shift of the SABR α parameter (with $\beta = 20\%$) which

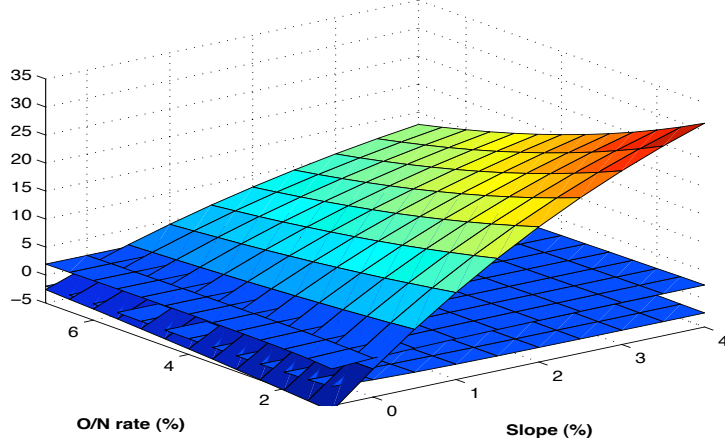


FIGURE 1. Differences between the physical/cash settle swaptions. Differences for different curve shapes. The constant color surfaces represent one SABR vega.

corresponds roughly to a 0.25% Black implied volatility shift. The goal is not to have an exact risk measure but an indication on the differences importance.

4.2. Hull-White. In the first part, the mean reversion factor is selected arbitrarily at 1%. The impact of the arbitrary parameter is analysed in a second step. The model volatility is calibrated to the market price of physical delivery swaptions with same maturity and strike. With that calibration we price cash-delivery swaption using Formula (1). The expectation is computed with a (one dimensional) numerical integration scheme. As a control, the physical delivery swaptions are repriced in the Hull-White model with the explicit formula recalled above and with the same numerical integration scheme used for the cash-delivery swaptions. For the physical delivery swaptions, the prices are exactly the same (more precisely: the numerical error is well below the precision displayed).

In Table 1, the prices for payer and receiver swaptions are reported. The difference between the physical and cash-settled swaptions are reported as a reference figure. The main objective is to analyze the difference between the market standard formula and a model price.

For out-of-the-money options, the difference is relatively small. Nevertheless for receiver it is of the same order of magnitude as the difference between market standard physical and cash swaptions. For in-the-money options the difference can be quite large. For far-in-the-money receiver, the difference can be up to 20 vegas (see our definition of vega above). The larger difference for receiver is understandable. The annuity has more impact for high rate; when all rates are zero, the annuities are simply a sum of accrual fractions. The impact of the different discounting annuities are more important for higher rate. Note also that the model impact does not disappear for very far in-the-money options, the contrary. The in-the-money options do not converge to a simple linear product like a swap, but to another contingent claim which can be viewed as a non-linear CMS. The model has also a larger impact on those options.

In the second set of tests, the same procedure is used with different mean reversion parameters. We used the parameters 0.1%, 1%, 2%, 5% and 10%. For each parameter, the same analysis is done. The results are reported in Table 2. As can be seen the results depend on the mean reversion in a non trivial way. The largest difference is up to 50 vega. For payers and low strikes, the difference can even change sign, depending on the mean reversion.

Price is only part of the analysis regarding derivatives. Another important part are the risks computed. Here we analyse the first order rate risk called delta which is the first derivative with

Strike	Delivery	Cash Mrkt	Del.-Mrkt	Cash HW	Mrkt-HW	Vega
Payer						
1.39	2201.87	2191.34	10.54	2187.37	3.97	1.13
2.89	1238.90	1232.97	5.93	1231.96	1.02	2.36
4.39	480.26	477.96	2.30	477.99	-0.03	3.62
5.89	159.88	159.11	0.76	159.22	-0.10	2.44
7.39	71.36	71.02	0.34	71.09	-0.07	1.32
8.89	39.30	39.11	0.19	39.15	-0.04	0.80
Receiver						
1.39	65.39	65.08	0.31	64.49	0.59	1.13
2.89	170.66	169.84	0.82	168.81	1.03	2.36
4.39	480.26	477.96	2.30	476.07	1.89	3.62
5.89	1228.12	1222.24	5.88	1217.59	4.65	2.44
7.39	2207.84	2197.27	10.56	2186.74	10.54	1.32
8.89	3244.02	3228.50	15.52	3208.48	20.02	0.80

Prices in basis points.

TABLE 1. Swaption prices (in basis points) for cash and physical delivery. Cash delivery with the market standard formula and with the Hull-White model ($a = 0.01$). The last column is the SABR vega (for 0.0001 increase of α). Market data as of 30 April 2010.

Strike	Mean reversion				
	0.1%	1%	2%	5%	10%
Payer					
1.39	5.50	3.97	2.26	-2.86	-11.32
2.89	2.04	1.02	-0.12	-3.50	-9.01
4.39	0.47	-0.03	-0.59	-2.26	-4.93
5.89	0.14	-0.10	-0.38	-1.19	-2.48
7.39	0.08	-0.07	-0.23	-0.72	-1.49
8.89	0.06	-0.04	-0.15	-0.48	-0.99
Receiver					
1.39	0.36	0.59	0.85	1.61	2.80
2.89	0.64	1.03	1.47	2.74	4.74
4.39	1.24	1.89	2.61	4.72	8.02
5.89	3.40	4.65	6.02	10.00	16.17
7.39	8.38	10.54	12.89	19.69	30.08
8.89	16.73	20.02	23.59	33.79	49.12

Prices in basis points.

TABLE 2. Swaption prices (in basis points) for cash-settled swaptions. difference between the market standard formula and the Hull-White model prices. Hull-White Model with different mean reversion parameter. Market data as of 30 April 2010.

respect to the zero-coupon rate (rescaled to a one basis point change). The delta is obtained with coherent dynamic of the calibrating instruments. By this we mean that the impact of the curve shift is applied to the physical delivery swaptions (by keeping the SABR parameters unchanged), those prices are used to calibrate a new Hull-White model with is used for the pricing of the cash-settled swaptions. This approach is also called *delta with recalibration*.

Some results are presented in Table 3. The swaption is a receiver swaption with a strike ATM+1.5%; its notional is 100m. The difference in delta for that particular swaption is above 200 EUR in total and bucketed delta are up to 600 EUR. Those figures are not very large as they correspond to less than the delta of 0.01 basis point. This would certainly be true if one was trading only those instruments. The structure of a lot of banks books is a large amount of (inter-bank) cash-settled swaption hedging an important amount of physical delivery swaptions (vanilla or exotic) for clients (corporate or retail). The notional amounts involved reach easily billions and the delta differences in thousands.

Tenor	Delta DSC			Delta FWD		
	Market	HW	Diff.	Market	HW	Diff.
5Y	-5719.63	-5703.09	16.54	40450.38	40175.56	-274.83
6Y	-248.12	-309.81	-61.69	-1727.81	-1640.72	87.09
7Y	-140.22	-192.83	-52.61	-2184.75	-2088.92	95.83
8Y	-67.00	-105.98	-38.98	-2481.73	-2380.44	101.28
9Y	-27.91	-49.28	-21.38	-2732.06	-2627.78	104.28
10Y	49.30	48.26	-1.04	-2953.67	-2849.37	104.30
11Y	116.80	139.34	22.54	-3187.40	-3084.44	102.96
12Y	164.67	212.51	47.84	-3364.43	-3264.93	99.50
13Y	150.09	224.32	74.23	-3462.60	-3367.65	94.95
14Y	112.26	214.99	102.73	-3541.70	-3452.44	89.25
15Y	-828.95	-677.95	151.00	-82320.01	-82929.33	-609.32
Total	-6438.70	-6199.52	239.18	-67505.77	-67510.47	-4.70
Zero-coupon delta in EUR.						

TABLE 3. Receiver swaption delta for cash delivery with the market standard formula and with the Hull-White model. Swaption notional of EUR 100m; strike of ATM+1.5%. Model with mean reversion parameter $a = 0.01$. Market data as of 30 April 2010.

In the last part for the Hull-White model, we analyse the difference for various rate curve shapes. The difference between the market formula price and the model price of the cash swaptions are computed for the same curves used in Figure 1. The swaption is the same receiver 5Yx10Y swaption with strike ATM+1.5%. The differences do not depend strongly on the curves. The differences are between 1.3 and 2.9 vega. The differences are the largest for low rates and steep curve. This is the type of curve prevailing in EUR at the moment of writing.

4.3. **G2++**. The model is calibrated to the market price of physical delivery swaptions. The calibration is done using the approximated formula described in Henrard (2010b) in a more general framework. The expectation required for the cash-settled option is computed with a (two dimensional) numerical integration scheme.

In the first set we calibrate to only one swaption of same maturity and strike. The mean reversion parameters are (arbitrarily) imposed (1% and 30%). The volatilities are calibrated on the market prices with the (arbitrary) constraint that the volatility for the first factor (mean reversion 1%) is four times the volatility of the second parameter. The correlation parameter impact is analyzed.

In Table 4, the results are presented for different correlation parameters with the same instruments as previously.

The correlation parameter has less impact than the mean reversion parameter of Hull-White but nevertheless a non-negligible one.

In the second set of tests, we calibrate the model to two swaptions for each pricing. We calibrate to the swaption of same tenor (10Y) and strike and to a swaption of tenor 1Y and same strike. The two volatilities are calibrated. In this way we calibrate short tenor and long tenor swaptions. The

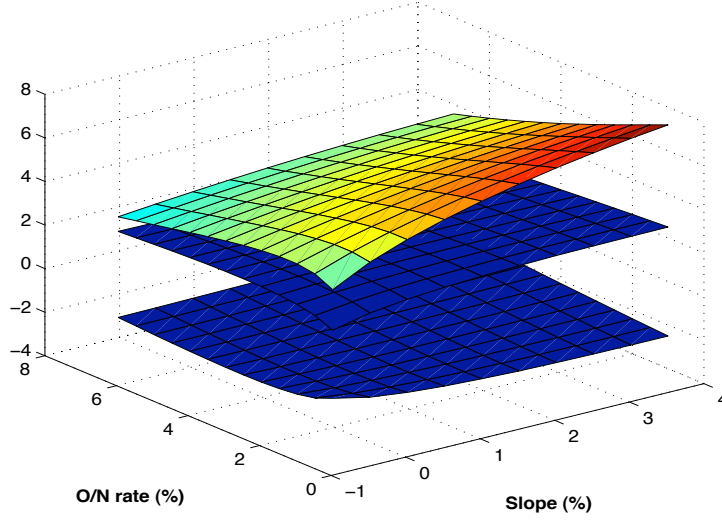


FIGURE 2. Differences between the market formula and calibrated Hull-White price. Differences for different curve shapes. The constant color surfaces represent one SABR vega.

Strike	Correlation				
	-90%	45%	0%	45%	90%
Payer					
1.39	6.02	5.17	4.15	3.36	2.64
2.89	2.97	2.34	1.75	1.20	0.59
4.39	0.89	0.30	-0.35	-0.31	-0.49
5.89	-0.97	-1.20	-1.38	-1.50	-1.55
7.39	-2.15	-1.79	-2.03	-4.48	-2.44
8.89	-0.67	-2.54	-2.91	-2.81	-1.62
Receiver					
1.39	3.34	0.74	1.49	1.39	2.94
2.89	1.36	1.61	1.93	1.98	2.20
4.39	1.98	1.66	1.96	2.27	2.89
5.89	2.17	2.91	3.00	4.18	5.15
7.39	5.96	7.41	8.80	9.85	11.04
8.89	13.61	15.92	17.88	19.80	21.25

Prices in basis points.

TABLE 4. G2++ model calibrated to 10Y swaptions. Mean reversion at 1% and 30%. Ratio between first and second factor volatility is four. Market data as of 30 April 2010.

results are also computed for several correlation parameters. By calibrating to two swaptions with a fixed mean reversion, we are not sure to always obtain a solution. In particular if the volatility tenor structure is not compatible to the selected mean reversions, one does not obtain an perfect fit. For that reason we display only the results for which a perfect fit was obtained. To be able to fit a larger set of parameter values, we took as mean reversion parameters 0.1% and 30%.

Strike	Correlation				
	-90%	45%	0%	45%	90%
Payer					
1.39	8.77				
2.89	3.91				
4.39		1.09	0.07	-0.09	0.03
5.89		-0.80	-1.31	-1.34	-1.57
7.39			-2.12	-1.97	-2.40
8.89		-2.96	-3.07	-2.73	-1.68
Receiver					
1.39	2.92				
2.89	1.01				
4.39		0.79	1.77	2.10	2.35
5.89		1.45	4.04	5.08	5.28
7.39			8.34	9.76	10.22
8.89		10.14	16.22	17.52	17.74

Prices in basis points.

TABLE 5. G2++ model calibrated to 10Y and 1Y swaptions. Results where perfect calibration was achieved. Market data as of 30 April 2010.

The results are not equal to the other calibration approach but the general picture is the same. For away from the money strikes, the difference can be very large.

4.4. LMM. We would like to calibrate to all the maturities up to the instrument maturity. In our case this means the swaptions 5Yx1Y up to 5Yx10Y. The calibration swaption have the same strike as the priced one. The parameters freedom of the LMM is larger than the one of the two other models. In the Libor Market Model two factors version we have two factors and two six month periods in each year. It means that we have four parameters to calibrate for each one year period. For each period we take the parameters weights as described below and calibrate a common multiplicative factor to obtained the market price.

We first analyze the multi-factor impact. The Libors used in the model construction are six month Libors. The volatilities γ_j are constructed with two components $\cos(\theta_j)$ and $\sin(\theta_j)$ and a norm. The angles θ_j are equally spaced between 0 and a final angle. The final angles are equally spaced between 0 and $\pi/2$. With 0, it is a one factor model, with $\pi/2$ it is a two factor model for with the changes of the six month rate in five years is independent of changes of the six month rate in 15 years. The displacement parameter is chosen to be 0.10. It represent a reasonable skew.

The results are displayed in Figure 6. Like for the previous models, large discrepancies appear for in-the-money instruments. Note that for in-the-money payers the difference is larger than the difference for the Hull-White and G2++ models while in the in-the-money receiver, the difference is smaller than for the Hull-White and G2++ models.

In the second part we analyze the skew impact. In the displaced diffusion models, the skew is represented by the displacement parameters. For each test we uses the same displacement for each Libor. We vary this parameter between the test from 0.05 to 1.00. The results are displayed in Figure 7. Like for the multi-factor LMM impact, the displacement impact is different from the one of Hull-White and G2++.

4.5. Comparison. In Figure 3 we summarise all the price data collected for the different models with market data from 30-Apr-2010. The vertical line represent the price in the market formula and the curved line the vegas for different strikes. The intervals represent the price range for different models. The plain lines are the payer swaptions and the dotted lines the receivers.

In Figures 4 and 5 the same results are presented for market data from 23-Apr-2009 and 30-Sep-2010. Obviously the quantitative results are different but the qualitative results are similar.

Strike	Angle				
	0	$\pi/8$	$\pi/4$	$3\pi/8$	$\pi/2$
Payer					
1.39	12.81	13.14	14.30	16.53	20.23
2.89	5.77	6.01	6.80	8.25	10.71
4.39	0.92	1.05	1.43	2.13	3.24
5.89	-0.38	-0.34	-0.15	0.23	0.80
7.38	-0.68	-0.65	-0.49	-0.14	0.33
8.88	-0.87	-0.83	-0.64	-0.37	0.06
Receiver					
1.39	-0.60	-0.65	-0.73	-0.73	-0.52
2.89	-0.15	-0.21	-0.42	-0.77	-1.14
4.39	1.24	1.13	0.72	-0.01	-1.22
5.89	5.09	4.84	4.14	2.82	0.57
7.39	9.68	9.30	8.15	6.03	2.32
8.89	15.47	14.94	13.27	10.04	4.51

Prices in basis points.

TABLE 6. Difference between Market standard formula and LMM prices. LMM model perfectly calibrated to swaptions with tenor from 1Y to 10Y. Results with different weights on the factors. Market data as of 30 April 2010.

Strike	Displacement		
	0.05	0.10	1.00
Payer			
1.39	19.53	14.29	10.46
2.89	8.65	6.80	5.24
4.39	2.03	1.43	0.83
5.89	0.30	-0.15	-0.21
7.39	0.01	-0.49	-0.13
8.89	-0.11	-0.64	0.03
Receiver			
1.39	0.42	-0.73	-0.43
2.89	0.25	-0.42	-0.53
4.39	0.93	0.73	0.75
5.89	3.87	4.14	5.44
7.39	7.05	8.15	11.99
8.89	10.79	13.27	21.24

Prices in basis points.

TABLE 7. Difference between Market standard formula and LMM prices. LMM model perfectly calibrated to swaptions with tenor from 1Y to 10Y. Results with different displacements. Market data as of 30 April 2010.

Note that for the 30-Sep-2010 data, the discrepancies are generally larger. This is not surprising as the rates were lower in a steep curve environment. The Figures 1 and 2.

5. CONCLUSION

The standard market formula for cash-settled swaptions is obtained by analogy to the Black formula. There is no modelling justification for the formula beyond the analogy. The formula can also be obtained by some *initial freeze* approximation.

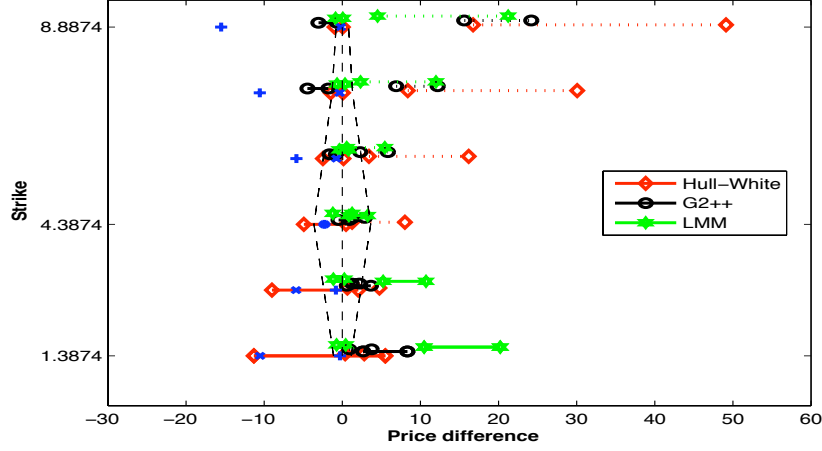


FIGURE 3. Differences for 5Yx10Y cash-settled swaptions. Plain lines represent payer swaptions and dotted line represent receiver swaptions. Market data as of 30-Apr-2010.

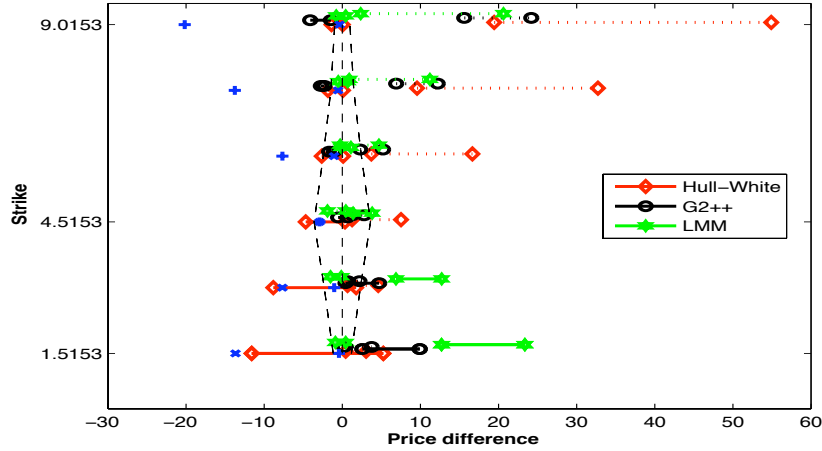


FIGURE 4. Differences for 5Yx10Y cash-settled swaptions. Plain lines represent payer swaptions and dotted line represent receiver swaptions. Market data as of 23-Apr-2009.

As shown in previous literature the standard smiles (Black or SABR) are not compatible with the formula. Together they create arbitrage opportunities.

In this note we analyse the formula from a different view point. The pricing of physical delivery swaption and cash-settled swaptions with their respective market formulas but with the same volatility is a standard market practice. That practice is analysed with several approaches

When tested with different one and two factor models, the price obtained by this approach can not be justified. Models calibrated to the relevant physical delivery swaptions show prices away from the market formula. For in-the-money options, the differences can be large.

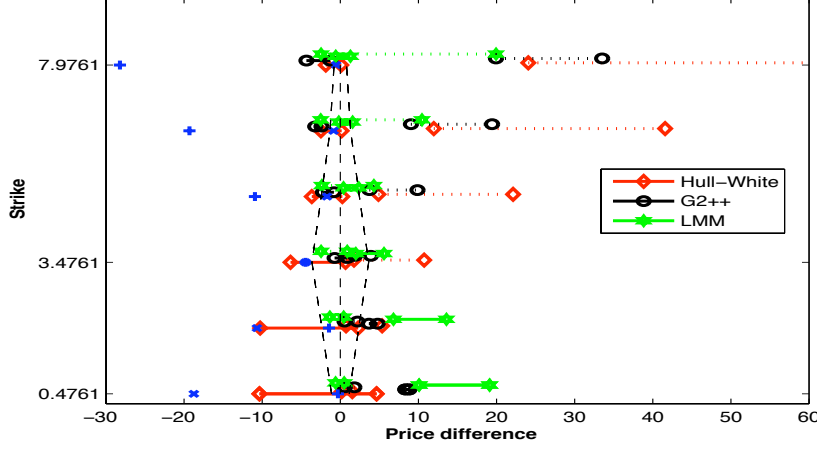


FIGURE 5. Differences for 5Yx10Y cash-settled swaptions. Plain lines represent payer swaptions and dotted line represent receiver swaptions. Market data as of 30-Sep-2010.

The differences are also influenced by non-calibrated factors. For the Hull-White model, the mean reversion has an important impact. For multi-factor models, the weights between the factor are important. The skew shape has also a non-negligible impact.

Those results question the wide usage of the market standard formula as only source of pricing for cash-settled and physical delivery swaptions.

APPENDIX A. APPROXIMATE FORMULA IN HULL-WHITE MODEL

The pay-off of the cash-settled swaptions can be viewed as an exotic CMS product. The technique used for the pricing of CMS like products in the gaussian HJM one factor model (Henrard (2008)) or multifactors (Hanton and Henrard (2010)) can be extended to the case analysed here.

For the Hull-White model, the technique idea is to take the expectation in Equation (1) and to replace the parts not easy to handel by their (second or third order) Taylor expansion. In our case the difficult part is the cash annuity and the swap rate exercise. Like in the CMS case, the exercise boundary is the same as in the physical delivery case described in Henrard (2003). The exercise boundary in the normally distributed random variable X is denoted κ and is given by

$$(3) \quad S_\theta(\kappa) = K.$$

The condition $S_\theta(X) < K$ becomes $X < \kappa$.

By opposition to the CMS case, the part to be approximated is not roughly linear in the random variable. The rate S_θ is roughly linear and the annuity is also roughly linear. The result is more parabola shape than a straight line. A third order approximation will be required to obtain a precise enough formula.

The pay-off expansion around the reference point X_0 is

$$C(S_\theta)(K - S_\theta) \sim U_0 + U_1(X - X_0) + \frac{1}{2}U_2(X - X_0)^2 + \frac{1}{3!}U_3(X - X_0)^3.$$

One could simply choose $X_0 = 0$ as a reference point. With that point, the approximation is globally good. For out-of-the-money options ($\kappa < 0$), it is better to concentrate the goodness of the approximation on the part of the interval that will actually be integrated. For receiver swaptions, this means the part below the exercise boundary. For those swaption the reference point we propose to use is $X_0 = \kappa$. For deep out-of-the-money options, this reduces significantly the approximation

error. In the example below, in the case of the lowest strike the error is decreased twenty fold with respect to the $X_0 = 0$ choice.

Theorem 1. *In the extended Vasicek model, the price of cash-settled receiver swaption is given to the third order by*

$$\begin{aligned} P(0, t_0) \mathbb{E} \left[\exp \left(-\tilde{\alpha}_0 X - \frac{1}{2} \tilde{\alpha}_0^2 \right) \left(U_0 + U_1(X - X_0) + \frac{1}{2} U_2(X - X_0)^2 + \frac{1}{3!} U_3(X - X_0)^3 \right) \right] \\ = \left(U_0 - U_1 \tilde{\alpha}_0 + \frac{1}{2} U_2(1 + \alpha_0^2) - \frac{1}{3!} U_3(\tilde{\alpha}_0^3 + 3\alpha_0) \right) N(\tilde{\kappa}) \\ + \left(-U_1 - \frac{1}{2} U_2(-2\tilde{\alpha}_0 + \tilde{\kappa}) + \frac{1}{3!} U_3(-3\tilde{\alpha}_0^2 + 3\tilde{\kappa}\tilde{\alpha}_0 - \tilde{\kappa}^2 - 2) \right) \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} \tilde{\kappa}^2 \right). \end{aligned}$$

where κ is given by Equation (3), U_i are the pay-off expansion coefficients, $\tilde{\kappa} = \kappa + \alpha_0$ and $\tilde{\alpha}_0 = \alpha_0 + X_0$.

The price of cash-settled payer swaption is given to the third order by

$$\begin{aligned} P(0, t_0) \mathbb{E} \left[\exp \left(-\tilde{\alpha}_0 X - \frac{1}{2} \tilde{\alpha}_0^2 \right) \left(U_0 + U_1(X - X_0) + \frac{1}{2} U_2(X - X_0)^2 + \frac{1}{3!} U_3(X - X_0)^3 \right) \right] \\ = - \left(U_0 - U_1 \tilde{\alpha}_0 + \frac{1}{2} U_2(1 + \alpha_0^2) - \frac{1}{3!} U_3(\tilde{\alpha}_0^3 + 3\alpha_0) \right) N(-\tilde{\kappa}) \\ - \left(U_1 - \frac{1}{2} U_2(2\tilde{\alpha}_0 - \tilde{\kappa}) + \frac{1}{3!} U_3(3\tilde{\alpha}_0^2 - 3\tilde{\kappa}\tilde{\alpha}_0 + \tilde{\kappa}^2 + 2) \right) \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} \tilde{\kappa}^2 \right). \end{aligned}$$

The quality of the approximation is presented in Table 8. The third order approximation gives precise results.

Strike	Integration	Approx. 2	Int-Appr. 2	Approx. 3	Int-Appr. 3
Payer					
-2.50	1817.53	1811.42	6.11	1817.86	-0.33
-1.50	1199.82	1195.24	4.59	1200.00	-0.18
-0.50	662.30	658.99	3.31	662.43	-0.14
0.00	458.80	455.73	3.07	458.95	-0.15
0.50	318.27	316.16	2.11	318.38	-0.10
1.50	177.44	176.29	1.15	177.50	-0.06
2.50	117.96	117.10	0.86	118.01	-0.05
Receiver					
-2.50	112.39	111.57	0.82	112.33	0.06
-1.50	175.30	174.21	1.10	175.23	0.07
-0.50	319.35	317.40	1.95	319.22	0.13
0.00	456.70	453.60	3.10	456.49	0.21
0.50	656.80	652.99	3.81	656.55	0.25
1.50	1195.97	1190.11	5.86	1195.61	0.37
2.50	1814.60	1805.32	9.28	1813.94	0.66
Prices in basis points.					

TABLE 8. Swaption prices for cash delivery. Hull-White price through numerical integration and approximation of order 2 and 3. Market data as of 31 July 2009.

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